EXAMINATION QUESTION PAPER

# GRA 41541
# Supply Chain Optimization with Mathematical Programming

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|---|---|
| **Start date:** | 06.02.2026 09.00 |
| **Finish date:** | 13.03.2026 12.00 |
| Weight: | 30% of GRA 4141 |
| No. of attachments files to question paper: | 0 |
| To be answered: | In groups of 1-3 students |
| Max no. of answer paper attachment files: | 1 |
| Allowed answer paper file types: | pdf |
| Allowed answer paper attachment file types: | zip, rar, etc. |

Attach AMPL, Excel, Python files etc. in zip / rar format.

All questions have equal weight.

# *BioNutrient*

*BioNutrient* is a supplier of fish feed that is used for the farming of cultivated salmon. The company has two production plants for feed, here called the Northern plant and the Southern plant.

There are hundreds of customers scattered all along the coast. For planning purposes, the customers are aggregated into 5 different sales districts.

The demand follows a highly seasonal pattern, as is evident from the following table that shows the estimated demand for each sales district for each quarter of the year.

Demand (tons):

| | Q1 (90 days) | Q2 (91 days) | Q3 (92 days) | Q4 (92 days) |
|---|---|---|---|---|
| District 1 | 4 000 | 19 000 | 34 000 | 15 000 |
| District 2 | 3 000 | 15 000 | 21 000 | 14 000 |
| District 3 | 7 000 | 24 000 | 33 000 | 25 000 |
| District 4 | 10 000 | 24 000 | 36 000 | 17 000 |
| District 5 | 15 000 | 38 000 | 48 000 | 27 000 |

To cope with the seasonal demand pattern, *BioNutrient* varies the number of shifts being used in the production. At both plants the maximum number of shifts is 5, and the minimum number of shifts is 3.

Production capacity Northern plant: 17 000 tons per shift per quarter.
Production capacity Southern plant: 15 000 tons per shift per quarter.

The production costs for each of the plants are as follows:
For the production capacities there is a fixed cost of NOK 4 mill. per shift per quarter.
However, for the 4$^{th}$ shift there is an additional overtime cost of NOK 1,5 mill. per quarter, and for the 5$^{th}$ shift the additional overtime cost is NOK 3,0 mill. per quarter.

The raw material is relatively expensive at the Southern plant because the raw materials need to be transported over a longer distance.
For the Northern plant the raw material costs are NOK 1 800 per ton, and for the Southern plant the raw material costs are NOK 2 000 per ton.

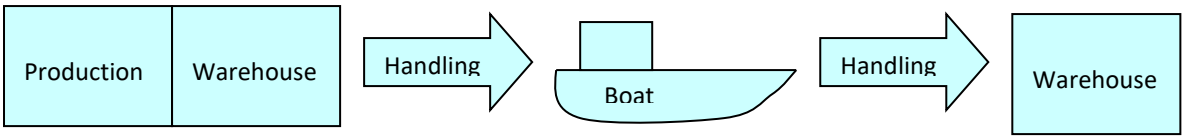
There is a warehouse at each of the plants, and in addition, *BioNutrient* has the possibility to rent a central warehouse located between the two plants. Renting the central warehouse costs NOK 7 mill. per year.

Storing finished product from one quarter to the next costs NOK 50 per ton. This cost does not depend on which warehouse is used.

Storage capacity at Northern plant: 8 000 tons.
Storage capacity at central warehouse: 16 000 tons.
Storage capacity at Southern plant: 9 500 tons.

Handling capacity Northern plant: 90 000 tons per quarter.
Handling capacity central warehouse: 60 000 tons per quarter.
Handling capacity Southern plant: 88 000 tons per quarter.

Handling here means the sum of the inbound and outbound flow for the warehouse. Handling includes loading and discharging of boats only; moving the products from a warehouse to a boat, and from a boat to a warehouse. It does not include the handling when moving products from the production to the warehouse at the same location. (See illustration.)




The products are transported by boat. There are eight different potential boats that can be used for this transport:

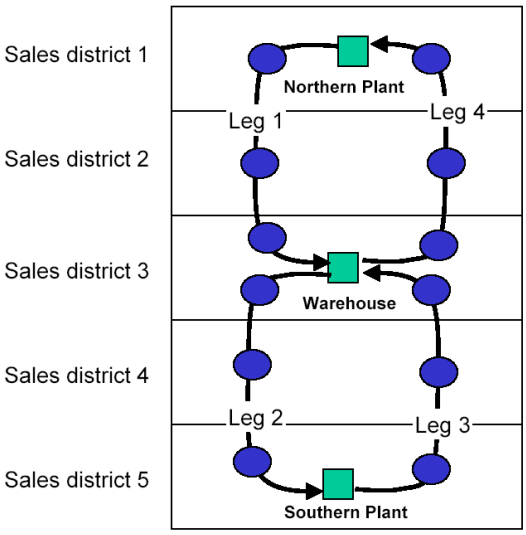
| Boat | Max load (tons) | Yearly costs (NOK) |
|---|---|---|
| Anna | 1 800 | 16 000 000 |
| Belinda | 1 700 | 15 400 000 |
| Carla | 1 600 | 14 800 000 |
| Diana | 1 400 | 13 800 000 |
| Emma | 1 200 | 12 000 000 |
| Fiona | 1 100 | 11 600 000 |
| Gunda | 1 000 | 10 800 000 |
| Huldra | 900 | 10 400 000 |

*BioNutrient* would prefer to rent as few of these boats as possible, in order to save transportation costs.

NB! Anna and Emma can also be rented for a part of the year (Q2 and Q3).
The cost for renting Anna in Q2 + Q3 is NOK 12 000 000.
The cost for renting Emma in Q2 + Q3 is NOK 9 400 000.

The boats are supposed to follow fixed routes, supplying the various customers in each district that is covered by the route. The transportation planners of *BioNutrient* have specified nine allowed routes (see the end of this file). Each route consists of a number of legs, where one leg goes from one warehouse to another warehouse. The route starts and ends up at the same warehouse.
The transportation planners have also specified for each leg which sales districts that are visited on the given leg.

Example of a route with four legs:




Leg 1:
Duration: 4 days
Districts: 1, 2, 3

Leg 2:
Duration: 4 days
Districts: 3, 4, 5

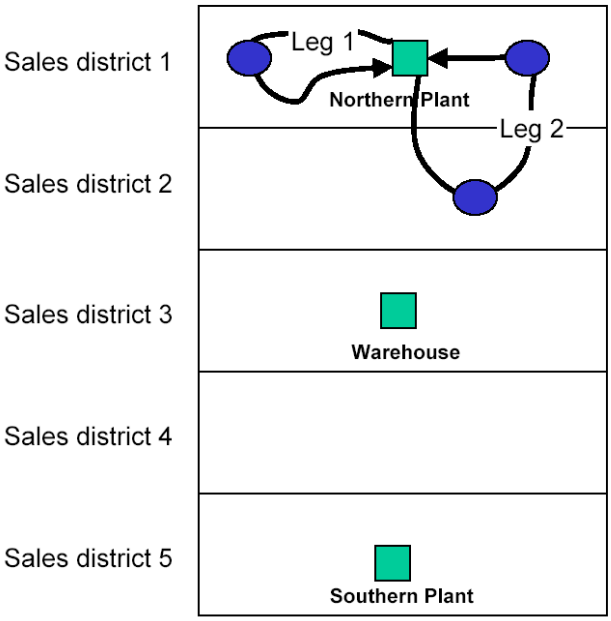
Leg 3:
Duration: 4 days
Districts: 5, 4, 3

Leg 4:
Duration: 4 days
Districts: 3, 2, 1

Total duration of the route:
16 days

As an example, assume that the boat Anna (max load 1 800 tons) is assigned to this route.
Anna can then deliver at most 4 × 1 800 = 7 200 tons for this 16-day period. That is, its delivery capacity will be 450 tons per day.
For Q1 (Jan-Feb-Mar), its total capacity will be 450 × 90 = 40 500 tons.

Example of a route with two legs:




Leg 1:
Duration: 2 days
Districts: 1

Leg 2:
Duration: 4 days
Districts: 2, 1

Total duration of the route:
6 days

If Anna is assigned to this route for Q1, it can deliver at most (1 800 + 1 800) = 3 600 tons for this 6-day period. That is, its delivery capacity will be 600 tons per day.
For Q1 (Jan-Feb-Mar), its total capacity will be 600 × 90 = 54 000 tons.

There can be no more than one boat per route (per quarter), and a boat can follow no more than one route per quarter. However, a given boat does not need to follow the same route for all quarters.

The company must pay a penalty cost of NOK 12 000 for each ton that is not delivered to the customers.

For simplicity, you can assume that the daily demand is constant *within* each quarter.
You can also assume that all boats have the same speed.
Specify which other assumptions you make.

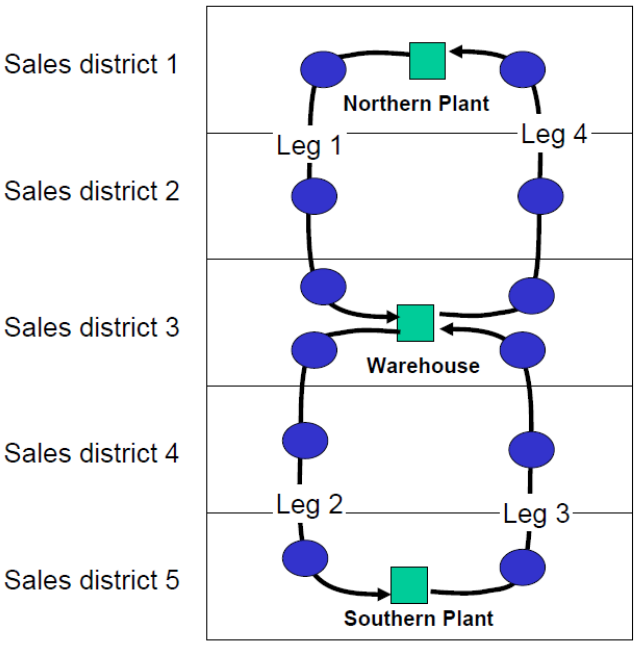
## Your task

*BioNutrient* needs to decide upon the production and distribution structure that will minimize total yearly costs for the coming year. The planners are overwhelmed by the complexity of the task, and you are asked to build an optimization model that can be applied to help the company answering the questions below.

1. Should the company rent the central warehouse?
2. How many shifts should be run at each plant in each quarter of the year?
3. How much should be produced at each plant in each quarter of the year?
4. How much should be transported between the warehouses in each quarter?
5. How much should be transported from each warehouse to each sales district in each quarter?
6. How much should be stored at each of the warehouses by the end of each quarter?
7. Which boats should be used in the transportation?
8. Which route should each boat follow in each quarter?
9. How much should be supplied by each boat to each sales district and to each warehouse each quarter?
10. What will be the total yearly costs for your plan? Specify the individual cost components (raw material costs, production capacity costs, etc...)
11. Imagine that your plan is rejected because it does not consider the uncertainty that will be present in a practical situation and its impact on customer service. What types of uncertainty could that be? How would you deal with this, that is, how would you take into account uncertainty and customer service?

## Allowed routes to choose from

### Route 1




Leg 1:
Duration: 4 days
Districts: 1, 2, 3

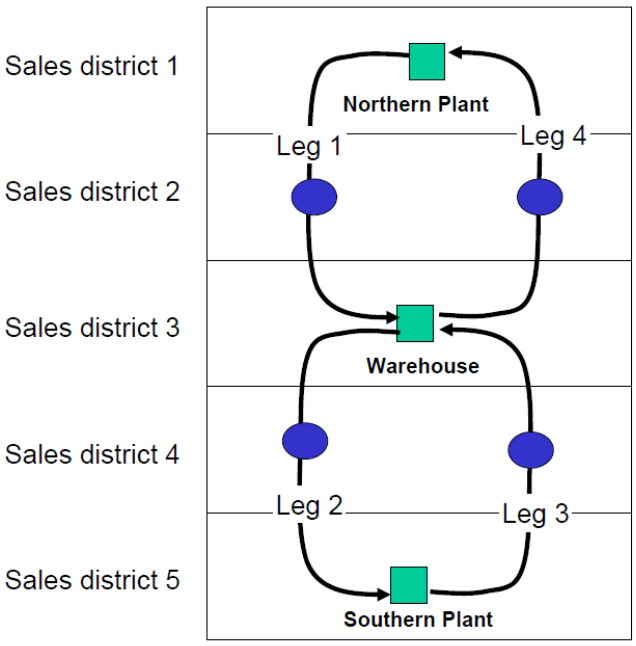
Leg 2:
Duration: 4 days
Districts: 3, 4, 5

Leg 3:
Duration: 4 days
Districts: 5, 4, 3

Leg 4:
Duration: 4 days
Districts: 3, 2, 1

Total duration of the route:
16 days

### Route 2




Leg 1:
Duration: 3 days
Districts: 2

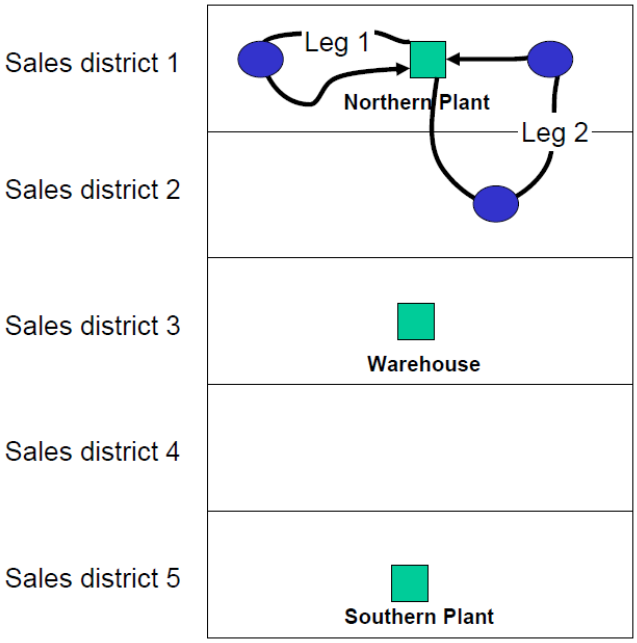
Leg 2:
Duration: 3 days
Districts: 4

Leg 3:
Duration: 3 days
Districts: 4

Leg 4:
Duration: 3 days
Districts: 2

Total duration of the route:
12 days

## Route 3

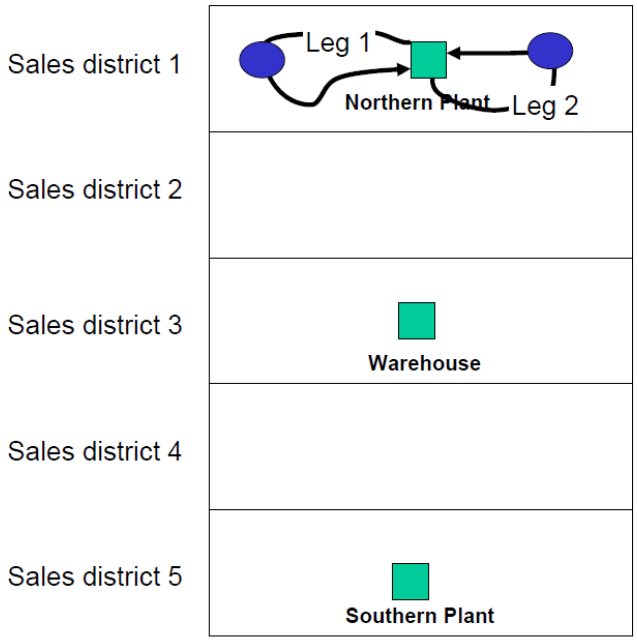



Leg 1:
Duration: 2 days
Districts: 1

Leg 2:
Duration: 4 days
Districts: 2, 1

Total duration of the route:
6 days

## Route 4

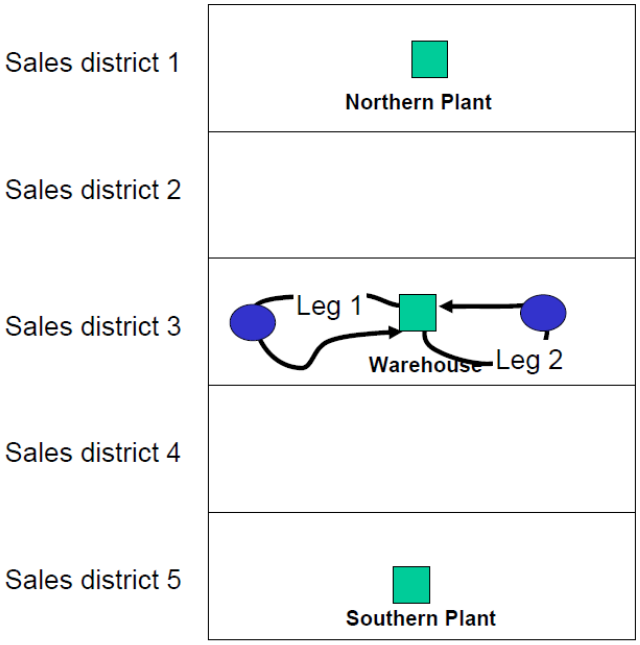



Leg 1:
Duration: 2 days
Districts: 1

Leg 2:
Duration: 2 days
Districts: 1

Total duration of the route:
4 days

## Route 5

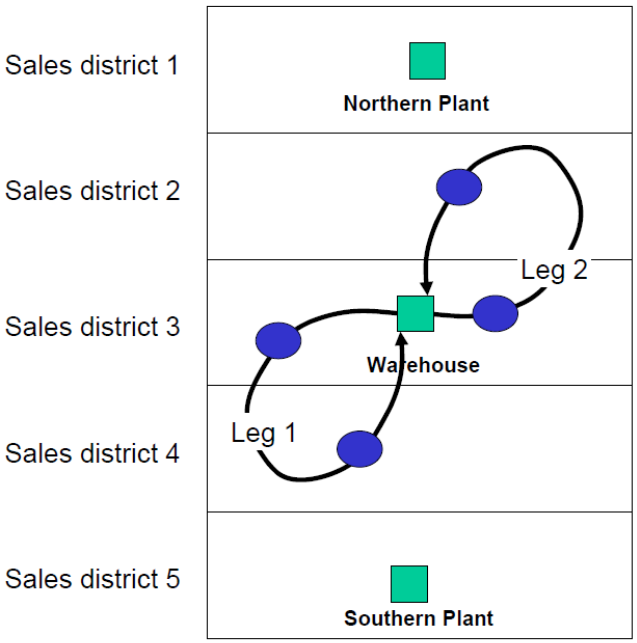



Leg 1:
Duration: 2 days
Districts: 3

Leg 2:
Duration: 2 days
Districts: 3

Total duration of the route:
4 days

## Route 6

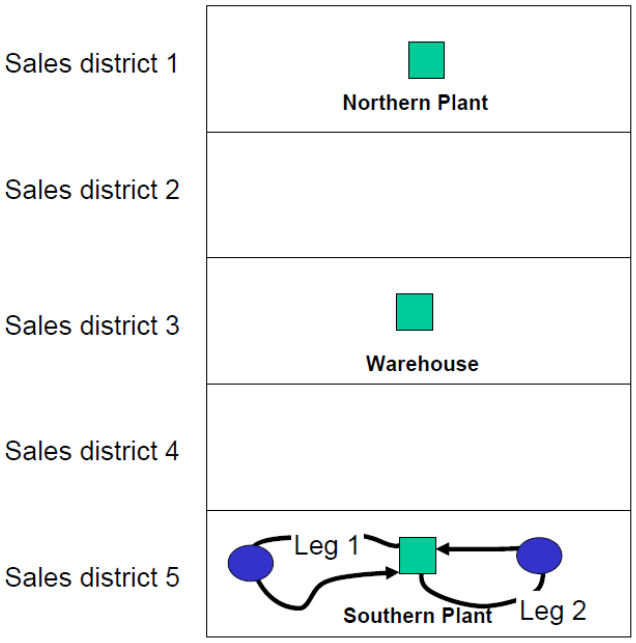



Leg 1:
Duration: 3 days
Districts: 3, 4

Leg 2:
Duration: 3 days
Districts: 3, 2

Total duration of the route:
6 days

## Route 7

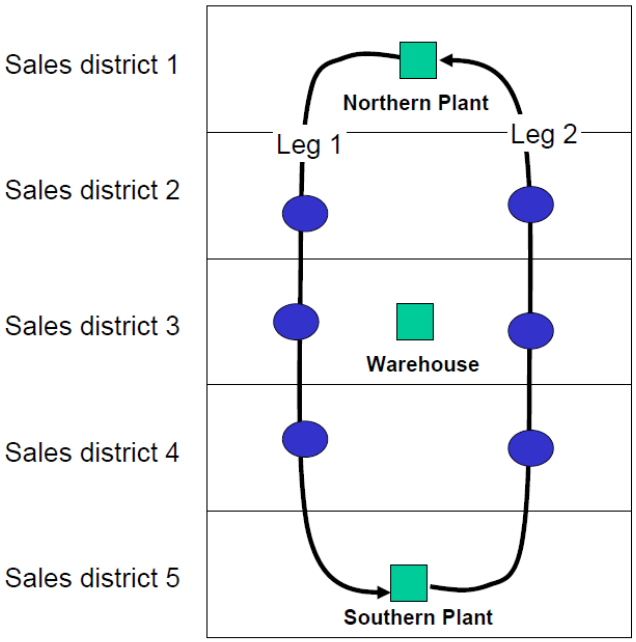



Leg 1:
Duration: 2 days
Districts: 5

Leg 2:
Duration: 3 days
Districts: 5

Total duration of the route:
5 days

## Route 8



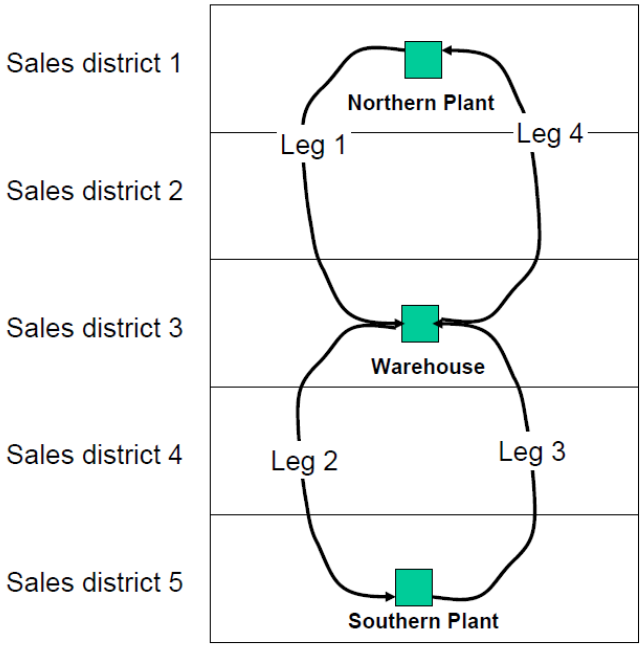

Leg 1:
Duration: 6 days
Districts: 2, 3, 4

Leg 2:
Duration: 6 days
Districts: 4, 3, 2

Total duration of the route:
12 days

(This route does not visit the central warehouse.)

## Route 9




Leg 1:
Duration: 2 days
Districts: None*

Leg 2:
Duration: 2 days
Districts: None*

Leg 3:
Duration: 2 days
Districts: None*

Leg 4:
Duration: 2 days
Districts: None*

Total duration of the route:
8 days

* Goes directly between the warehouses only